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ORIGINAL ARTICLE

Genetic Architecture of N-Terminal Pro-B-Type Natriuretic Peptide in a Multiancestry Study Population

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BACKGROUND: NPs (natriuretic peptides) are bioactive hormones crucial for regulating blood pressure, glucose homeostasis, and lipid metabolism. Despite the high heritability of circulating NP levels, the genetic determinants of NP regulation, particularly across ancestries and sexes, remain poorly understood. The objective of the current study was to identify genetic variants associated with NT-proBNP (N-terminal pro-B-type NP) levels in a multiancestry study population.

METHODS: Whole genome sequencing and array-based data from 81 213 individuals without heart failure were analyzed from the Trans-Omics for Precision Medicine cohorts, UK Biobank, All of Us Research Program, and REGARDS (Reasons for Geographic and Racial Differences in Stroke) study to identify common, rare, and structural variants associated with NT-proBNP levels. The main outcome of the study was log-transformed and standardized NT-proBNP levels. Genetic associations with NT-proBNP were examined, followed by gene prioritization, transcriptome-wide association studies, colocalization, and rare variant analyses.

RESULTS: Nine novel loci and 3 previously reported loci were identified to be associated with NT-proBNP levels. Novel structural variants were detected across 12 loci. Similar effect sizes were observed for both common and rare variants. Key genes such as *BAG3* (10q26.11) and *SLC39A8* (4q24) were identified through gene prioritization, with prior animal models supporting their therapeutic relevance. Rare variant analysis identified 6 masks with significant associations, specifically noncoding masks, suggesting regulatory modulation of NT-proBNP.

CONCLUSIONS: This study identifies novel common, rare, and structural variants associated with NT-proBNP levels, highlighting the contribution of both coding and regulatory noncoding variation. These findings advance our understanding of the genetic architecture of NT-proBNP and may inform future cardiometabolic therapeutic strategies.

Key Words: atrial natriuretic factor ■ blood pressure ■ cardiovascular diseases ■ genetics ■ lipid metabolism

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NPs (natriuretic peptides) are cardiometabolic hormones that regulate various physiological functions, including vasodilation, natriuresis, lipid metabolism, and glucose utilization, all of which contribute to maintaining cardiovascular and metabolic health.[1–6] NPs, particularly NT-proBNP (N-terminal pro-B-type NP), are key biomarkers in cardiovascular disease and serve as inclusion criteria and primary end points in major heart failure (HF) trials.[7–9] Understanding the genetic determinants of NP levels is essential for improving clinical care and optimizing HF treatment strategies. The genetic contribution to circulating NP levels is substantial, accounting for over a third of interindividual variability.[10]

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Nonstandard Abbreviations and Acronyms

AoURP	All of Us Research Program
ARIC	Atherosclerosis Risk in Communities
BP	blood pressure
CAGE	Cap Analysis of Gene Expression
CHS	Cardiovascular Health Study
eQTL	expression quantitative trait loci
FHS	Framingham Heart Study
FUMA	functional mapping and annotation
GWAS	genome-wide association study
HF	heart failure
LD	linkage disequilibrium
LDSC	linkage disequilibrium score regression
MAGMA	Multi-Marker Analysis of Genomic Annotation
MESA	Multi-Ethnic Study of Atherosclerosis
NP	natriuretic peptide
NPX	Normalized Protein Expression
NT-proBNP	N-terminal pro-B-type natriuretic peptide
QC	quality control
REGARDS	Reasons for Geographic and Racial Differences in Stroke
SNP	single-nucleotide polymorphism
SV	structural variant
TOPMed	Trans-Omics for Precision Medicine
WGS	whole genome sequencing

Previous genome-wide association studies (GWASs) have identified significant loci influencing NP levels, providing important insights into their genetic regulation. Prior studies found that the *MTHFR-CLCN6-NPPA-NPPB* gene cluster was a critical determinant of NT-proBNP levels.[11] Similarly, other studies have highlighted the relevance of atrial NP in blood pressure (BP) regulation.[12] A GWAS performed using data from the Jackson Heart Study identified 2 loci, *NPPB* and *KLKB1*, and these loci were also found to be associated with cardiovascular traits such as systolic BP and left ventricular mass.[13] However, prior GWAS have been limited by small sample sizes and lack of ancestral diversity.[11–13]

Ancestry and sex differences in NP levels have been well established.[14–20] Females have higher NP levels, and individuals of African ancestry have lower NP levels compared with their counterparts.[14–20] Ancestry- and sex-specific GWASs may provide novel insights into the differences in the genetic underpinnings for circulating NP levels. Furthermore, a multiethnic GWAS has the potential to identify additional loci that influence NP levels across populations.

In this study, we conducted GWAS for NT-proBNP in the Trans-Omics for Precision Medicine (TOPMed) cohorts, the REGARDS study (Reasons for Geographic and Racial Differences in Stroke), the All of Us Research Program (AoURP), and the UK Biobank. Subsequently, a meta-analysis of the GWAS results was performed, making this study of 81 213 individuals the largest to date in examining the genetic architecture of NT-proBNP. This study was conducted in a large multiethnic study population and was aimed (1) to estimate the heritability of circulating NPs, (2) to identify genetic loci associated with circulating NP levels through overall, and ancestry- and sex-stratified GWAS, and (3) to explore the broader genetic architecture of circulating NPs, including the contribution of rare and structural variants (SVs).

METHODS

The authors will make their data, analytical methods, and study materials available to other researchers on reasonable request.

Study Cohorts

The current study included individuals from the TOPMed cohorts, the REGARDS study, AoURP, and the UK Biobank with NT-proBNP measurements and genetic data available. The TOPMed cohorts included the ARIC study (Atherosclerosis Risk in Communities), CHS (Cardiovascular Health Study), the FHS (Framingham Heart Study), and the MESA (Multi-Ethnic Study of Atherosclerosis), which underwent whole genome sequencing (WGS) under the TOPMed Initiative. Details of each of these cohorts have been provided in the Supplemental Methods. The studies were approved by the institutional review boards at each of the participating centers. All participants provided written informed consent. The AoURP enrolled individuals aged ≥18 years from >340 sites across the United States between 2017 and 2022.[21] Individuals with WGS data were included in the study.

The UK Biobank is a prospective population cohort that was initiated in 2006 and has enrolled 502 493 participants aged between 40 and 69 years from 22 centers across the United Kingdom.[22] The REGARDS study recruited 30 239 Black and White individuals aged ≥45 years in 2003 to 2007 across the United States. Genotype array data from the UK Biobank and the REGARDS study were used in this study.

The current study excluded those with suspected HF based on age-specific NT-proBNP levels (<75 years: ≥125 pg/mL and ≥75 years: ≥450 pg/mL were excluded).[23] In the UK Biobank, NT-proBNP levels were quantified in Normalized Protein Expression (NPX).[24,25] Therefore, NT-proBNP-based cutoffs for HF could not be applied directly in the UK Biobank. Hence, participants were excluded if they had prevalent HF based on *International Classification of Diseases* codes and self-report. Due to the strong correlation between NPX values and protein levels measured by the ELISA-based assay, we used the NPX values generated from the Olink Panel for this analysis.

Genotyping and Quality Control

In the TOPMed cohorts, WGS with a goal of 30× coverage was performed on the Illumina HiSeq X instruments at 8

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sequencing centers using polymerase chain reaction–free library preparation kits.[26] Samples were centrally aligned to the GRCh38 human genome reference build using BWA-MEM at the TOPMed Informatics Research Center.[27] Joint variant calling was performed by the TOPMed Informatics Research Center using the GotCloud pipeline. These variant calling software are developed by the TOPMed Informatics Research Center and can be accessed through GitHub[28] (http://github.com/atks/vt, http://github.com/hyunminkang/apigenome, and https://github.com/statgen/topmed_variant_calling). Quality control (QC) at the sequencing, variant, and sample levels was performed by the TOPMed Informatics Research Center. Sequencing quality criteria comprised excluding samples with estimated DNA contamination of >12% and average coverage <15×. Variant QC was done by first calculating the Mendelian consistency scores based on duplicates and known familial relatedness, and training a support vector machine classifier for previously discovered variants and the variants identified with Mendelian inconsistencies. Additional QC steps included an excessive heterozygosity filter (Hardy-Weinberg disequilibrium P value of $<1\times10^{-6}$ in the direction of heterozygosity) and a Mendelian discordance filter (≥3 Mendelian or duplicate discordance observed from the samples). Sample-level QC included the removal of samples with pedigree errors, reported and genotypic sex discrepancies, and discordance with prior genotyping array data. The variant call format files generated from the above protocol were split by consent groups for distribution to approved dbGaP users.

In the AoURP, standardized laboratory protocols, pipelines, and QC steps were used across all of the All of Us Genome Centers.[29] WGS in the AoURP involved polymerase chain reaction–free library construction using the Illumina Kapa HyperPrep kit.[29] The Illumina NovaSeq 6000 was utilized for sequencing. The Illumina DRAGEN pipeline was used for mapping, aligning, variant calling, and QC. The GRCh38 reference genome was used for alignment. QC steps included a mean coverage of ≥30×, genome coverage of ≥90% at 20×, hereditary disease risk gene coverage of ≥95% at 20×, contamination ≤1%, and concordance with array data.[29]

In the UK Biobank, genotyping was performed on 2 closely related genotyping arrays: the UK BiLEVE Axiom array (n=49 950) and the UK Axiom array (n=438 427).[30] The 2 arrays shared ≈95% of the markers.[30] Genotype calling was performed by Affymetrix using a custom pipeline.[30] Variant-level QC incorporated testing for batch effect, plate effect, departures from the Hardy-Weinberg equilibrium, sex effect, array effect, and discordance across controls.[30] Sample-level QC included excess heterozygosity, genotyping rate, and sex discordance.[30] Imputation was performed centrally by the UK Biobank. Genotypic data were phased and lifted over from GRCh37 to GRCh38.[31] Imputation was performed on the TOPMed R2 reference panel using Minimac4.[26,32] Variants with poor imputation score were excluded (imputation quality metric $R^2<0.1$).

In the REGARDS study, genotyping was conducted on the Illumina Infinium Multi-Ethnic AMR/AFR Extended BeadChip array. QC measures included the removal of samples with internal duplicates, sex discordance, and a high proportion of missing variants (>5%) and variants with MAF <5%, nonbiallelic variants, strand inconsistencies, violating the Hardy-Weinberg equilibrium, and variant call rate <10%.[33] Imputation was performed using the TOPMed reference panel (version R2) on the National Heart, Lung, and Blood Institute TOPMed Imputation server using Minimac4.[26,32,34] Postimputation QC was performed to exclude variants with imputation quality <0.3.

NT-proBNP Measurements

NT-proBNP was measured on the Roche Cobas e411 analyzer using Elecys proBNP II immunoassay (Roche Diagnostics, Indianapolis, IN) in ARIC. In MESA and CHS, NT-proBNP was measured on the Roche Elecsys 2010 instrument (Roche Diagnostics). In FHS, NT-proBNP was estimated on a Luminex xMAP platform (Sigma-Aldrich, St. Louis, MO). The coefficient of variation for these instruments has been previously reported.[35–38] In the AoURP, NT-proBNP levels were measured using analyzers available at the institution where the participants presented. NT-proBNP levels were log-transformed to account for the skewed distribution.

The UK Biobank used the Olink Explore 3072 assay to measure NT-proBNP levels among other proteins, which reports protein abundance as NPX values.[24,25] These are log2-transformed and normalized internally by Olink, which accounts for technical variations. Given the log2-transformation, no additional transformation was performed.

In the REGARDS study, NT-proBNP levels were measured using the Roche Elecsys 2010 immunoassay.[15,39]

The method used to address data censoring for NT-proBNP measurements across different cohorts is explained in detail in the Supplemental Methods.

Given the differences in NT-proBNP levels, the rank-based inverse normal transformation approach was used to account for potential assay censoring and nonnormal trait distributions. This step was performed across 4 cohorts and used these standardized units for genetic analysis. A detailed explanation of the method used can be found in Supplemental Methods.

Single Variant Analysis

The genome-wide analyses for variants with minor allele count >20 were conducted in each cohort individually. The association of each variant with standardized log-transformed NT-proBNP was analyzed using SAIGE-QT linear mixed models with kinship adjustment on the Encore platform (https://encore.sph.umich.edu/) in TOPMed cohorts.[40] Similarly, SAIQE-QT linear mixed models were used for analysis in the UK Biobank. The mixed linear model-based GWAS tool called fastGWA in the REGARDS in unrelated individuals from the AoURP was used for analysis.[41,42] These models were adjusted for age, age,[2] sex, and the first 10 principal components of genetic ancestry (principal components 1–10). To account for multiple testing, a Bonferroni correction accounting for the number of variants analyzed was used.

Using the summary statistics from the 4 cohorts, a fixed-effects meta-analysis was conducted using METAL.[43] We used a 1-stage meta-analysis framework using an inverse weighted fixed-effects model. This approach weights effect size estimates by the inverse of their standard errors. Correction for genomic control was applied at the meta-analysis level to account for any residual inflation.

Considering the sex- and ancestry-based differences in circulating NP levels, sex- and ancestry-specific GWASs were conducted. Similar to the primary analysis, GWASs for

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NT-proBNP were conducted for males, females, European ancestry individuals, and African ancestry individuals separately in each cohort. The resulting GWAS summary statistics were meta-analyzed using METAL to obtain the meta-analyzed summary statistics for all subgroups. The meta-analysis was conducted to account for study-specific variability arising from the use of different analytical tools, all of which have been validated in previous large-scale GWAS efforts.

Standard QC and Genomic Inflation assessment were performed to test the potential inflation due to population stratification. Quantile-quantile plots were generated for all cohorts and meta results, and genomic control inflation factors (λGC) were calculated and reported (Table S1; Figure S1). In addition, linkage disequilibrium (LD) score regression (LDSC) was used to calculate the intercept and quantify residual inflation attributable to stratification versus polygenicity, and the results are reported.[44]

The variants identified by genome-wide significance ($P \leq 5\times10^{-8}$) were clumped with $r^2>0.6$ within a window size of 500 kb to identify 52 significant independent single-nucleotide polymorphisms (SNPs). A Bonferroni cutoff accounting for the number of independent variants identified in the discovery was used to determine statistical significance. To identify lead SNPs associated with NT-proBNP among the variants passing the Bonferroni threshold, LD clumping was performed using $r^2>0.1$ and a window size of 500 kb using PLINK, version 1.90b3.45 64-bit (January 13, 2017).[45] The 1000 Genomes reference panel,[46] including all ancestry groups, was used for clumping. LD clumping was also performed for subgroups (European, African, Male, and Female) using the Bonferroni threshold from the primary analysis.

Fine-Mapping

To identify putative causal variants within genome-wide loci, we performed fine-mapping using the sum of single effects regression using summary statistics, implemented in the susieR package. We predefined 12 genomic regions based on genome-wide significant loci from the meta-analysis and conducted a sum of single effects regression using summary statistics separately for European and African ancestry cohorts using ancestry-specific GWAS summary statistics and LD matrices derived from the 1000 Genomes Project.[47] Z scores were calculated from effect size estimates and standard errors. LD matrices were symmetrized and adjusted to be near positive-definite to ensure model stability. For the European ancestry analysis, we used L=10 to allow for multiple causal variants per region. For the African ancestry analysis, L=1 and L=3 were used to reflect the smaller sample size and corresponding statistical power. Posterior inclusion probabilities were estimated, and 95% credible sets were constructed for each locus. Regions with missing LD or summary statistics were excluded from analysis. Given the smaller sample size in the African ancestry GWAS, the power to detect multiple signals may be limited, necessitating conservative assumptions in fine-mapping.

Functional Mapping and Annotation

Each locus was annotated using functional mapping and annotation (FUMA) for expression quantitative trait loci (eQTL) associations and gene expression analyses.[48] FUMA integrates data from 14 biological repositories and tools to prioritize lead SNPs and conduct eQTL associations and gene expression analyses. The SNP2GENE function maps the lead variants to genes. The GENE2FUNC function annotates the genes identified in the SNP2GENE function with biological functions.

Gene Prioritization

To prioritize potentially causal genes in the loci identified to be associated with NT-proBNP, several analyses were conducted, which include the following.

1. Genes identified based on gene-based tests performed using Multi-Marker Analysis of Genomic Annotation (MAGMA; Table S9).
2. Genes identified through transcriptome-wide association study of the left ventricle or the left atrial appendage using FUSION (Tables S13 and S14).
3. Genes crossing PP.H4≥80% based on eQTL colocalization analysis using the left ventricle tissue–specific eQTL data set from GTEx, version 8 (Table S15).
4. Genes near each independent variant identified by genomic significance ($P \leq 5\times10^{-8}$; Data Set S6).
5. Genes identified through polygenic priority scores[49] analysis with MAGMA enrichment Z score ≥1 (Data Set S7).
6. Genes mapped to tissue-specific eQTL variants specific to the left ventricle or the left atrial appendage on GTEx, version 8, where PP.H4≥75%[50] (Table S18).
7. Genes that are linked to Hi-C chromatin states and showing interaction based on false discovery rate $<1\times10^{-6}$ (Table S19).[51]
8. Genes mapped to exonic SNPs of genome-wide independently significant variants (Table S20).
9. Genes linked to variants that are shown to have RegulomeDB scores ≤2[52] (Data Set S8).
10. Genes with a probability of being loss-of-function intolerance ≥0.90 based on the Exome Aggregation Consortium browser extracted as part of the FUMA GWAS tool[53] (Table S22).
11. Genes linked to 95% credible variants identified through ancestry-stratified fine-mapping (Data Set S9).
12. Genes that are protein-coding with knockout mouse-related models available and linked to cardiovascular ontology terms (Data Set S10).

For each of the abovementioned criteria, genes meeting the criteria were given a point. A sum of the points was calculated to determine the gene prioritization score, ranging from 0 to 14. Genes with the highest scores within each locus were reported as prioritized genes for that locus. A detailed explanation of the abovementioned methods can be found in Supplemental Methods.

SV Analysis

SV analysis was conducted in the TOPMed cohorts using the TOPMed SV freeze 1.0 call set. Briefly, the Parliament2 pipeline was used to make SV calls for each sample separately.[54] This pipeline integrates SV calls from 6 programs, including BreakDancer, BreakSeq, CNVnator, Delly, Lumpy, and Manta. Survivor[55] was used to merge the SV calls across the 6 programs, and SVTyper[56] was used to filter the calls. muCNV was used to examine the sample genotypes for each variant.[57]

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Using a 500-kb window size around each cytogenic band containing a lead variant, SVs in the region were identified. Using SAIGE-QT mixed models adjusted for age, age,^{2} sex, study, and the first 10 principal components, genome-wide associations between NT-proBNP levels and SVs were assessed. Subsequently, results were reported focusing on the prioritized genomic loci identified in the original analysis. SVs crossing the Bonferroni threshold (0.05/12 loci) were reported. To identify the gene containing the SV, AnnotSV was used to annotate the significant SVs.[58]

In the AoURP, SV analysis was conducted using GATK-SV. A detailed explanation of the steps of QC and variant calling has been described elsewhere.[59,60] SVs identified in 3477 unrelated individuals with minor allele count >20 were included in the association analysis. Linear regression models were used for GWAS, adjusting for age, age,^{2} sex, and the first 10 principal components. Results were prioritized according to loci reported in the original meta-analysis and subsequently reported. SVs located within the identified loci were queried, and those crossing the Bonferroni significance threshold (0.05/12 loci) were reported. Genes encompassing the SV regions were identified using AnnotSV.[58]

Heritability Estimation

LDSC regression was used to estimate the heritability of NT-proBNP.[44] The LDSC method of estimating heritability requires the GWAS summary statistics and LD scores. In this method, it is assumed that the estimated effect of an SNP combines the effect of all SNPs in LD with that SNP. SNPs that are in LD with a larger number of SNPs are more likely to tag a causal variant compared with an SNP that is in LD with a smaller number of SNPs. Therefore, SNPs with a larger LD score are more likely to have a larger effect size compared with SNPs with a lower LD score. LDSC method of estimating heritability regresses the SNP effect size obtained from the GWAS summary statistics on the LD score of the SNP. The relationship of these 2 variables provides an estimate of SNP-based heritability based on the number of SNPs included in the analysis. This analysis was performed using precurated LD baseline model estimates.[61] The LD scores generated using the 1000 Genomes reference panel for European ancestry were used for this analysis.

Rare Variant Analysis

Rare variant analysis was performed across 4 cohorts using Regenie, version 3.1.1, and Regenie, version 3.4, with the former used for null model generation and the latter used for association testing. Coding masks included putative loss-of-function variants, missense variants, splice sites, and a combined putative loss-of-function+missense mask. Noncoding masks included intronic, intergenic, downstream, upstream, and untranslated region variants. In addition, we incorporated cardiac-specific promoter and enhancer regions annotated from Cap Analysis of Gene Expression (CAGE) and DNase I Hypersensitivity Site data sets to capture regulatory elements active in heart tissue. All variant annotations were derived using the FAVOR, version 20211005, database and extracted using in-house scripts.[62] Results were meta-analyzed using METAL, and associations surpassing the Bonferroni-corrected significance threshold were reported. A detailed description of the variant masks generation and rare variant analysis methodology was provided in the Supplemental Methods.

RESULTS

This study included 81 213 individuals with NT-proBNP measurements and genetic data. The median age was 59 (50–65) years, 54.5% were females, and 81.9% were individuals of European ancestry (Table 1). The median NT-proBNP values were 53.0 (25.5–105.8), 119.0 (49.0–288.0), and 83.6 (37.7–194.8) pg/mL in the TOPMed cohorts, AoURP, and REGARDS study, respectively. The median NT-proBNP levels, presented in NPX, in the UK Biobank were −0.09 (−0.68 to 0.54).

After GWAS, the quality of the results was assessed using multiple metrics. Quantile-quantile plots generated showed appropriate deviation from the null, consistent with polygenicity without evidence of systemic inflation (Figure S1). λGC assessed across different cohorts, and subgroups ranged between 1 and 1.5, with higher values observed in the UK Biobank due to the large sample size and statistical power (Table S1). LDSC regression on meta-analysis yielded an intercept of 0.98 (SE=0.0072) and λGC of 1.01, suggesting a modest inflation, which is primarily driven by polygenicity rather than unaccounted population structure.

Common Variant Associations

From the meta-analyzed GWAS using a 1-stage framework and LD-based clumping, 52 lead variants were identified representing 12 loci across 10 chromosomes (Data Set S2; Figures 1 and 2). The 12 loci identified were 1p36.13 (*SRARP* and *HSPB7*), 1p36.22 (*NPPB*), 4q24 (*SLC39A8*), 5q31.2 (*WNT8A*), 6p21.2 (*CDKN1A*), 7q32.1 (*FLNC*), 8q23.1 (*ZFPM2*), 10q22.2 (*SYNPO2L*), 10q26.11 (*BAG3*), 12q21.33 (*POC1B*), 15q26.3 (*IGF1R*), and 18q21.31 (*NEDD4L*; Table 2). These loci included previously reported loci (*NPPB-CLCN6-MTHFR*, *POC1B*, and *SLC39A8* loci) associated with NT-proBNP.[11,12,64] There were 9 novel loci identified in this study, which included 1p36.13 (*SRARP* and *HSPB7*), 5q31.2 (*WNT8A*), 6p21.2 (*CDKN1A*), 7q32.1 (*FLNC*), 8q23.1 (*ZFPM2*), 10q22.2 (*SYNPO2L*), 10q26.11 (*BAG3*), 15q26.3 (*IGF1R*), and 18q21.31 (*NEDD4L*). Given the use of a 1-stage meta-analysis design, measures of heterogeneity (Cochran Q and I^2) across the cohorts were calculated and reported for all genome-wide significant loci, as presented in Table 2.

Sex-stratified GWAS results for NT-proBNP have been presented in Figure 3 and Data Set S3. In comparing the associations of the SNPs with NT-proBNP in the overall GWAS in both sexes, the direction of the effect estimates for all lead SNPs was consistent in males and females (Figure 4). However, only 33

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Table 1. Characteristics of the Studied Population Samples

	AoURP	REGARDS	TOPMed	UK Biobank
N	9640	3776	18 000	49 797
Age, y	61.0 (51.0 to 71.0)	67.0 (59.0 to 75.0)	58.0 (50.0 to 67.0)	58.0 (50.0 to 63.0)
Female (%)	5454 (56.6)	1861 (49.3)	9887 (54.9)	27 048 (54.3)
Risk factors				
Systolic blood pressure, mm Hg	128.0 (116.3 to 140.3)	139.0 (125.0 to 153.0)	120.0 (109.0 to 134.5)	136.5 (124.5 to 149.5)
Diastolic blood pressure, mm Hg	77.7 (70.3 to 85.7)	83.0 (75.0 to 91.0)	72.0 (65.0 to 79.0)	82.0 (75.0 to 88.5)
Body mass index	30.4 (26.0 to 35.9)	28.7 (25.2 to 32.9)	26.9 (24.0 to 30.4)	26.7 (24.2 to 29.9)
Diabetes	3701 (38.5)	779 (20.7)	1690 (9.6)	3519 (7.4)
Hypertension	8249 (85.9)	2577 (68.4)	7635 (43.2)	26 228 (53.9)
Medication use				
Lipid-lowering medication	5364 (55.8)	1206 (32.3)	1901 (10.7)	8894 (17.9)
Blood pressure medication	7865 (81.8)	2235 (61.4)	5393 (30.5)	10 653 (21.4)
NT-proBNP levels (pg/mL or NPX)				
Levels	119.0 (49.0 to 288.0)	83.6 (37.7 to 194.8)	53.0 (25.5 to 105.8)	−0.01 (−0.8 to 0.8)
n (%) below LOD	137 (1.4)	38 (1.0)	659 (3.7)	*
Ancestry				
European	5988 (62.1)	2245 (59.5)	12 577 (69.8)	46 466 (93.3)
African	1801 (18.7)	1531 (40.5)	2745 (15.3)	1136 (2.3)
Others	1851 (19.2)		2678 (14.9)	2195 (4.4)

AoURP indicates All of Us Research Program; LOD, limit of detection; NPX, Normalized Protein Expression; NT-proBNP, N-terminal pro-B-type natriuretic peptide; REGARDS, Reasons for Geographic and Racial Differences in Stroke; and TOPMed, Trans-Omics for Precision Medicine.

*Not mentioned because of the NPX values of the UK Biobank.

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variants in males and 39 variants in females reached genome-wide significance ($P \leq 5 \times 10^{-8}$; Data Set S3). Similarly, ancestry-stratified GWASs were conducted for NT-proBNP (Data Set S4; Figure 3). Significant variants identified in the overall GWAS showed consistent direction of effects in both ancestry groups, with 48 variants significant in individuals of European ancestry and 2 variants significant in those of African ancestry (Data Set S4). Figure 4 illustrates the directionality of effects for SNPs that passed the Bonferroni correction in both sex- and ancestry-stratified analyses, highlighting consistent effect directions across these subgroups. Table S6 presents the lead SNPs identified for each stratified category analyzed, along with their respective R^2 values in comparison to the lead SNPs reported in Table 2. Most of the lead SNPs identified here are either correlated with variants or identical to those in Table 2. Interestingly, the 6q22.31 locus (SLC35F1) was identified in European Ancestry GWAS but not observed in multiancestry meta-analysis.

Identification of Credible Variant Sets Using Fine-Mapping From Ancestry-Stratified GWAS Summary Statistics

Fine-mapping was conducted for the 12 genome-wide significant loci associated with NT-proBNP, identified through meta-analysis, using ancestry-stratified GWAS summary statistics. For each ancestry group, 95% credible sets were generated to identify variants with high posterior inclusion probabilities. In the African ancestry–specific fine-mapping, 2 loci, 1p36.22 and 6p21.2, were resolved: 1 with 4 variants and the other with 3 variants in their respective credible sets. Similarly, in the European ancestry-based fine-mapping, 5 loci (1p36.22,4q24, 7q32.1, 10p26.11, and 15q26.3) have 2 credible sets, 3 loci with 3 credible variants (10q22.2, 12q21.33, and 1p36.13), and the remaining loci with >3 credible variants. The most complex locus was 6p21.2 with a total of 9 credible variants. These findings are summarized in Table S7, with full ancestry-specific credible sets detailed.

Functional Annotation and Gene Mapping

Functional annotation of the genome-wide significant SNPs associated with NT-proBNP levels was performed using FUMA to understand the function of the SNPs in the genomic loci. Variants significantly associated with NT-proBNP were predominantly enriched in the intronic 814 (53.3%) and intergenic 336 (22.0%) regions, with 65 (4.3%) variants located in exonic regions (Figure S2). There were 72 variants with a combined annotation-dependent depletion score of ≥12.37 (indicating suggestive deleteriousness), and 118 variants had RegulomeDB scores (category 1-2) likely to affect binding.

Using the 3 approaches (positional mapping, eQTL mapping, and chromatin interaction mapping) implemented

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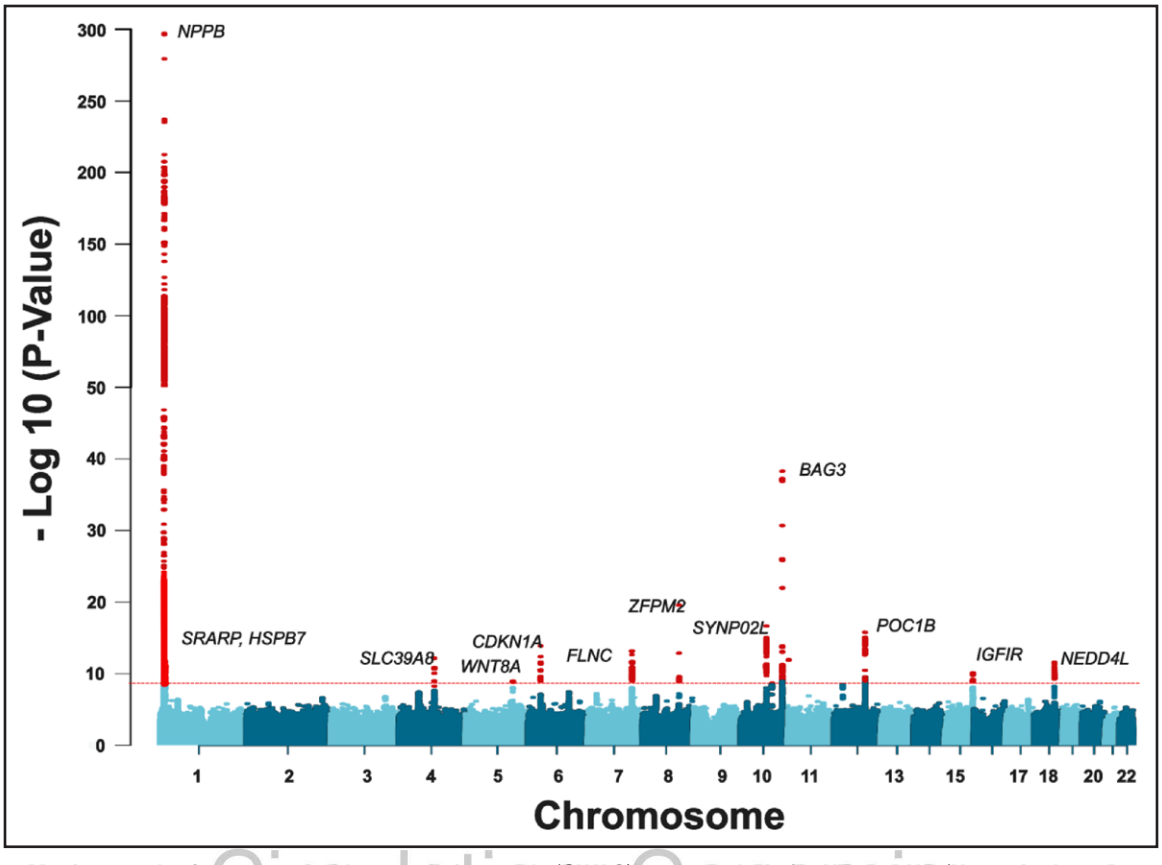

Figure 1. Manhattan plot for genome-wide association study (GWAS) meta-analysis for NT-proBNP (N-terminal pro-B-type natriuretic peptide). Manhattan plot showing significant variants identified for NT-proBNP when performing GWAS on 81 213 individuals across 4 cohorts.

The *x* axis depicts the chromosome number, and the *y* axis depicts the –log10 *P* values for the association of each individual genetic variant with NT-proBNP levels. The Bonferroni-corrected threshold ($P<1.43\times10^{-9}$) is represented by the dotted line. Independent genome-wide significant loci have been presented in red. Genes linked to lead single-nucleotide polymorphism were reported for each locus that crossed the Bonferroni cutoff.

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in FUMA, the significant SNPs associated with NT-proBNP levels were mapped to 168 genes, of which 75 were protein-coding genes. Among the 75 protein-coding genes, 21, 3, and 68 genes were mapped by positional mapping, eQTL mapping, and chromatin interaction mapping, respectively. There were 58 (77.3%) and 17 (22.6%) genes mapped by a single strategy and ≥2 strategies, respectively (Data Set S5; Figure 5). Chromatin interaction plots have been depicted in Figure S3.

Gene-Based Analysis

Gene-based association analysis was conducted using MAGMA, wherein SNPs were first mapped to genes, and then, the association of the genes with NT-proBNP levels was tested.[65] This analysis revealed 29 genes that were significantly associated with NT-proBNP levels after correcting for multiple testing ($P<0.05/19\,226$; Table S9; Figure S4). Not surprisingly, the top gene identified was *NPPB* ($P2.39\times10^{-15}$). Among the 29 genes identified, there were 14 genes that were not identified through the FUMA-based mapping methods.

Tissue Enrichment and Pathway Analysis

MAGMA tissue–specific gene expression analysis was performed using GTEx, version 8, gene expression data.[50] This analysis revealed that NT-proBNP-related gene expression was enriched in heart tissue, specifically the left atrial appendage and the left ventricle (Tables S10 and S11; Figure S5). Pathway enrichment analysis using MAGMA was performed utilizing curated pathway and GeneOntology term data from the Molecular Signatures Database.[66] This study identified 11 enriched GeneOntology terms, of which 1 was a molecular function term, 4 were cellular component terms, and the remaining were canonical pathways (Table S12). The significant GeneOntology terms were related to the sarcomere, extracellular matrix structure, and ion channel activity (Figure S6).

Transcriptome-Wide Association Study

A transcriptome-wide association study was performed using FUSION[67] and identified 17 unique genes associated with NT-proBNP levels in left ventricle tissue.

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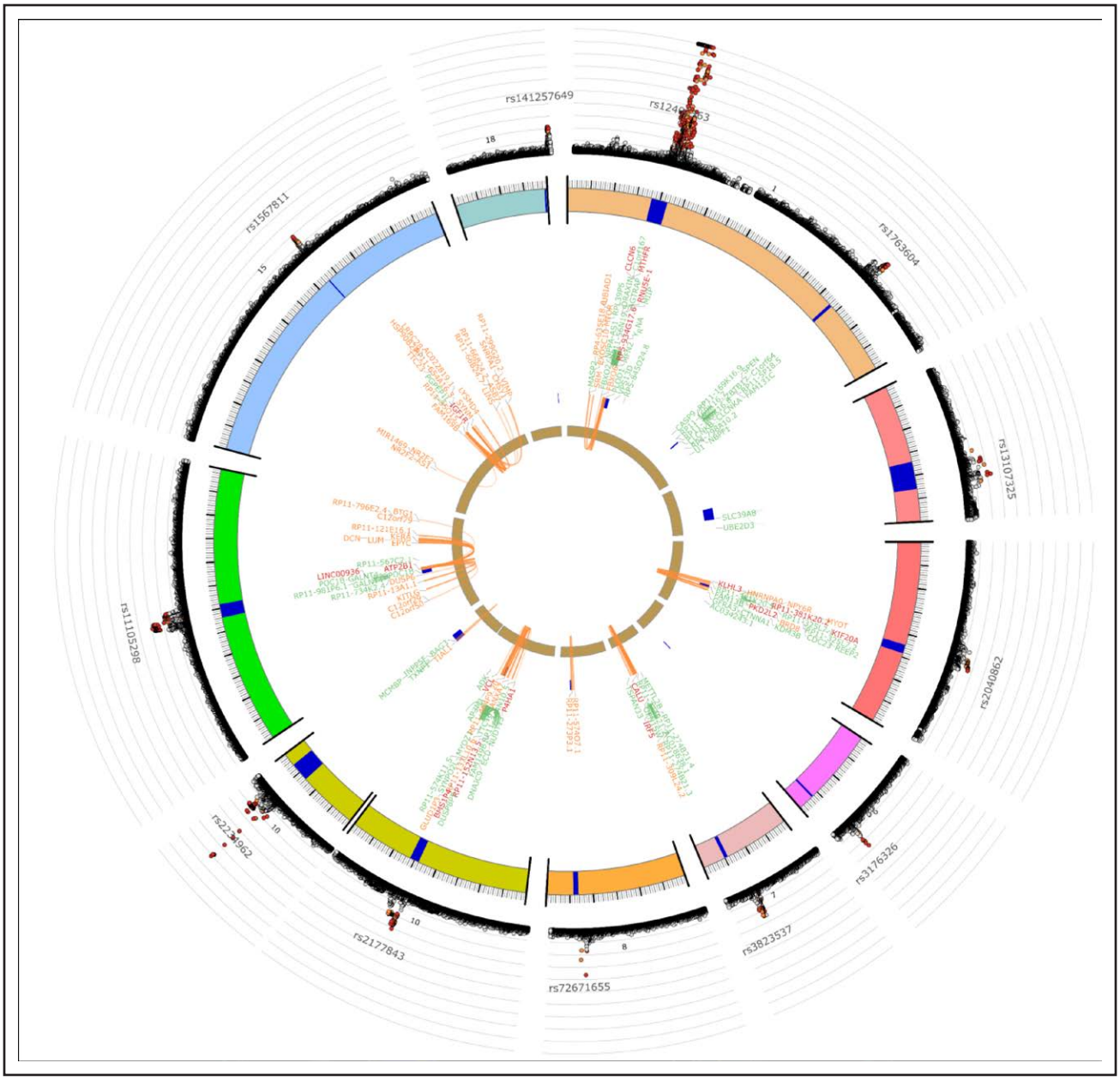

Figure 2. Circos plot for genome-wide association study meta-analysis for NT-proBNP (N-terminal pro-B-type natriuretic peptide). The figure depicts variants that were significantly associated with NT-proBNP at the Bonferroni-corrected threshold ($P<1.43\times10^{-9}$) for genome-wide significance at the 12 genomic loci identified in the study.
The inner region of the plot depicts the mapped genes by chromatin interactions (in green) and expression quantitative trait loci (in orange) or both (in red).

Among these, 3 genes, including *VGLL2* (chr6), *FADS1* (chr11), *PRKCA* (chr17), and PI4KAP2 (chr22), were located outside of the 12 genomic loci identified through GWAS (Table S13; Figure 6). A similar analysis was performed in atrial appendage tissue, yielding 13 genes associated with NT-proBNP levels, of which 1 gene was outside the identified GWAS loci (Table S14; Figure S7). Conditional and joint analyses were performed to examine if the GWAS and transcriptome-wide association study associations were independent of each other (Figures S8 and S9).

eQTL Colocalization in Heart Tissue

Colocalization analysis was conducted to examine whether the same genetic variants contributed to the associations between GWAS signals and eQTLs in the left ventricular tissue. There were 10 genes in the left ventricle tissue with strong or moderate colocalized signals. Genes such as *AC113382.1* (PP.H4=98.2%; nSNPs=1176), *FAM13B* (PP.H4=97.4%; nSNPs=442), and *IGF1R* (PP.H4=79.6%; nSNPs=702) showed high posterior probability of colocalization, suggesting shared effects between

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Table 2. Significant Loci and Their Lead SNPs Associated With NT-proBNP

Cytogenic band	Region	Lead SNP	Nearest gene	Effect allele	β	SE	*P* value	I^2 (%)	Q-Stat
1p36.13	chr1:15900000–20100000	1:16013277:C:G	*SRARP* and *HSPB7*	G	0.04	0.01	2.37×10^{-12}	63.2	8.2
1p36.22	chr1:9100000–12500000	1:11861186:A:G	*NPPB*	G	0.22	0.01	6.45×10^{-298}	97.2	105.3
4q24	chr4:100100000–106700000	4:102267552:C:T	*SLC39A8*	T	0.08	0.01	6.78×10^{-13}	93.8	48.4
5q31.2	chr5:136900000–140100000	5:138084300:C:T	*WNT8A*	T	0.04	0.01	1.05×10^{-09}	30.3	4.3
6p21.2	chr6:36600000–40500000	6:36679512:G:A	*CDKN1A*	A	−0.05	0.01	1.34×10^{-14}	67.7	9.3
7q32.1	chr7:127500000–129600000	7:128839286:G:A	*FLNC*	A	0.07	0.01	7.81×10^{-14}	70.4	10.1
8q23.1	chr8:105100000–109500000	8:105335669:T:A	*ZFPM2*	A	0.13	0.01	2.58×10^{-20}	87.4	23.8
10q22.2	chr10:73100000–75900000	10:73650119:C:T	*SYNPO2L*	T	−0.06	0.01	2.17×10^{-17}	62.8	8.1
10q26.11	chr10:117300000–119900000	10:119670121:T:C	*BAG3*	T	−0.09	0.01	5.14×10^{-39}	89.8	29.3
12q21.33	chr12:88600000–92200000	12:89482366:T:C	*POC1B*	T	0.05	0.01	1.68×10^{-16}	92.5	39.8
15q26.3	chr15:98000000–101991189	15:98738604:C:G	*IGF1R*	C	0.04	0.01	7.20×10^{-11}	14.7	3.5
18q21.31	chr18:56200000–58600000	18:58292226:A:ACTC	*NEDD4L*	A	0.04	0.01	2.50×10^{-12}	89.6	28.7

Significant variants were identified and associated with inverse rank normal transformed NT-proBNP were clumped as independent variants using a *P* value threshold of 1.43×10^{-9}, r2 value of 0.1, and a window size of 500 kb from the sentinel SNP using PLINK2, taking the 1000 Genomes data set as reference. Lead variants were defined as variants that were independent and significant per locus identified above and annotated using the SNPnexus[63] tool. SNPnexus takes the variants list as input and reports the necessary annotations along with their cytogenic band, which defines the locus. This table identifies the cytogenic band, region, and lead SNP (lowest *P* value) for each locus. Heterogeneity was assessed using inverse-variance weighted fixed-effects meta-analysis implemented in METAL. Cytogenic band indicates the genomic locus in which the gene is located; I^2, proportion of total variance due to heterogeneity; Lead SNP, independent variant with lowest *P* value within each locus; Q-Stat, Cochran Q statistics to test for between ancestry heterogeneity; NT-proBNP, N-terminal pro-B-type natriuretic peptide; Region, location of the locus region; SE, SE of the effect size of the SNP; SNP, single-nucleotide polymorphism; and β, effect size of the SNP.

GWAS and eQTLs in this tissue (PP.H4≥75%). *NPPB* showed moderate colocalization evidence (PP.H4=47.8%; nSNPs=643; Table S15; Figure S10).

Gene Prioritization

Several in silico analyses were undertaken to prioritize the likely causal gene within each locus among the 75 genes containing protein-coding variants identified through gene mapping strategies. A gene prioritization score was calculated using 14 criteria (described in the Supplemental Methods). Based on the gene prioritization score, the *MFN2* gene in the 1p36.22 locus, the *HSPB7* gene in the 1p36.13 locus, the *SLC39A8* gene in the 4q24 locus, the *CDKN1A* gene in the 6p21.2 locus, the *ZFPM2* gene in the 8q23.1 locus, the *BAG3* gene in the 10q26.11 locus, and the *NEDD4L* gene in the 18q21.31 locus were prioritized (Figure 7). The *MFN2* gene was prioritized with a high gene prioritization score of 9.

SV Analysis

SV analysis performed in the TOPMed cohorts identified 4 SVs associated with NT-proBNP levels after correction for multiple testing (0.004 [0.05/12]). These variants included 2 deletions and 2 duplications in chromosomes 1 and 12 of the lead loci. In the AoURP, SV analysis yielded 13 variants across 8 loci after accounting for multiple testing. Among the 14 variants identified, there were 8 breakends, 3 deletions, 1 duplication, and an insertion each. 1p36.22 and 15q26.3 are loci with a significant number of SVs (Table 3).

Heritability Analysis for NT-proBNP

Using the LDSC partitioned heritability method, the estimated SNP-based heritability of circulating NT-proBNP levels was estimated to be 5.1%.

Rare Variant Analysis

Using rare variant aggregation testing across coding, noncoding, and regulatory masks, we identified several associations with NT-proBNP levels. The strongest signal was observed at the *NPPB* locus using the combined putative loss-of-function+missense and missense mask, showing a positive association (β=0.33; SE=0.02; $P=4.24\times10^{-65}$). A stronger signal was also detected in the *NPPB* promoter region annotated from CAGE data (β=0.20; SE=0.01; $P=2.33\times10^{-44}$), supporting a key role for noncoding regulatory variants in modulating NT-proBNP levels. Additional significant associations in regulatory regions included the *CLCN6* promoter (β=0.08; SE=0.02; $P=1.38\times10^{-8}$) and the *ARHGAP42* enhancer region (β=−0.13; SE=0.02; $P=9.0\times10^{-7}$). The intronic mask of *RNF151* and the missense mask of *KIAA1257* were also shown to have a strong signal. Full association results are presented in Table 4, with a Manhattan plot highlighting these signals shown in Figure S11.

DISCUSSION

This study represents the largest effort to date to comprehensively characterize the genetic architecture of NPs, utilizing data from over 80 000 individuals from

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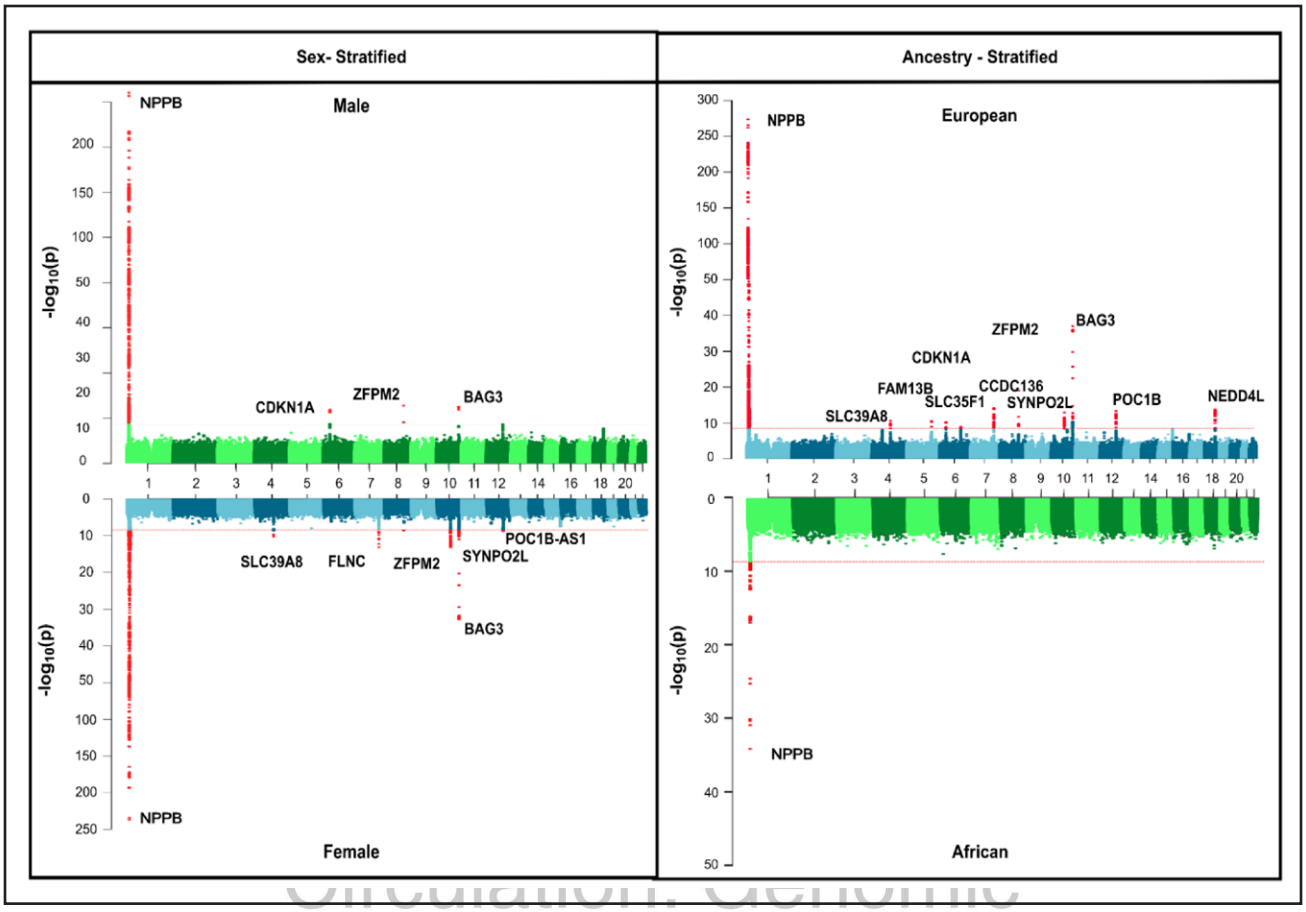

Figure 3. Manhattan plot for genome-wide association study meta-analysis for NT-proBNP (N-terminal pro-B-type natriuretic peptide) stratified by sex and ancestry.
Left and **right**, Sex-stratified and ancestry-stratified genome-wide associations for NT-proBNP levels, respectively. **Left**, Males are depicted on the top, and females are depicted on the bottom. **Right**, European ancestry individuals are depicted on the top, and African ancestry individuals are depicted on the **bottom**. The *x* axis depicts the chromosome number, and the *y* axis depicts the −log10 *P* values for the association of each genetic variant with NT-proBNP levels. The Bonferroni-corrected threshold ($P<1.43\times10^{-9}$) is represented by the dotted line. Independent genome-wide significant loci have been presented in red.

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diverse ancestries. Utilizing data from the TOPMed cohorts, REGARDS study, AoURP, and UK Biobank, the current study replicated 3 previously identified genomic loci and identified 9 novel genomic loci. Notably, this study found that the associations of lead SNPs identified as significant in the overall GWAS were consistent across sex and ancestry subgroups although they did not reach statistical significance within these subgroups. This suggests that the genetic determinants identified in the overall GWAS may be robust across different demographic groups, highlighting their potential relevance. In addition to common variants, SVs in the genome were associated with circulating NT-proBNP levels, highlighting the complexity of the genetic influences on NP regulation. Ancestry-stratified fine-mapping was performed at genome-wide significant loci for NT-proBNP. This analysis highlights several loci with consistent associations across ancestries, specifically at 1p36.22 and 6p21.2. At these loci, African ancestry-based fine-mapping resulted in smaller credible sets, illustrating the utility of diverse LD patterns in refining association signals, although the analysis was limited by a smaller sample size within the African ancestry group. These findings underscore the added value of multi-ancestry analysis in fine-mapping and emphasize the importance of including diverse populations in genetic studies.

Using a rigorous gene prioritization pipeline encompassing 14 criteria, the current study identified the *MFN2*, *HSPB7*, *SLC39A8*, *CDKN1A*, *ZFPM2*, *BAG3*, and *NEDD4L* genes as the prioritized genes within their respective loci. Notably, our lead variants replicated previously reported NT-proBNP associations in *NPPB* and *SLC39A8*.[11,12,64] Although our bioinformatics pipeline prioritized *MFN2* at the 1p36.22 locus based on integrated scoring, this locus is functionally anchored by *NPPB*, which encodes the precursor of BNP. Considering the role of the *NPPB* gene in the production of BNP, the association of the *NPPB* locus with circulating NT-proBNP levels is not surprising. This study validates

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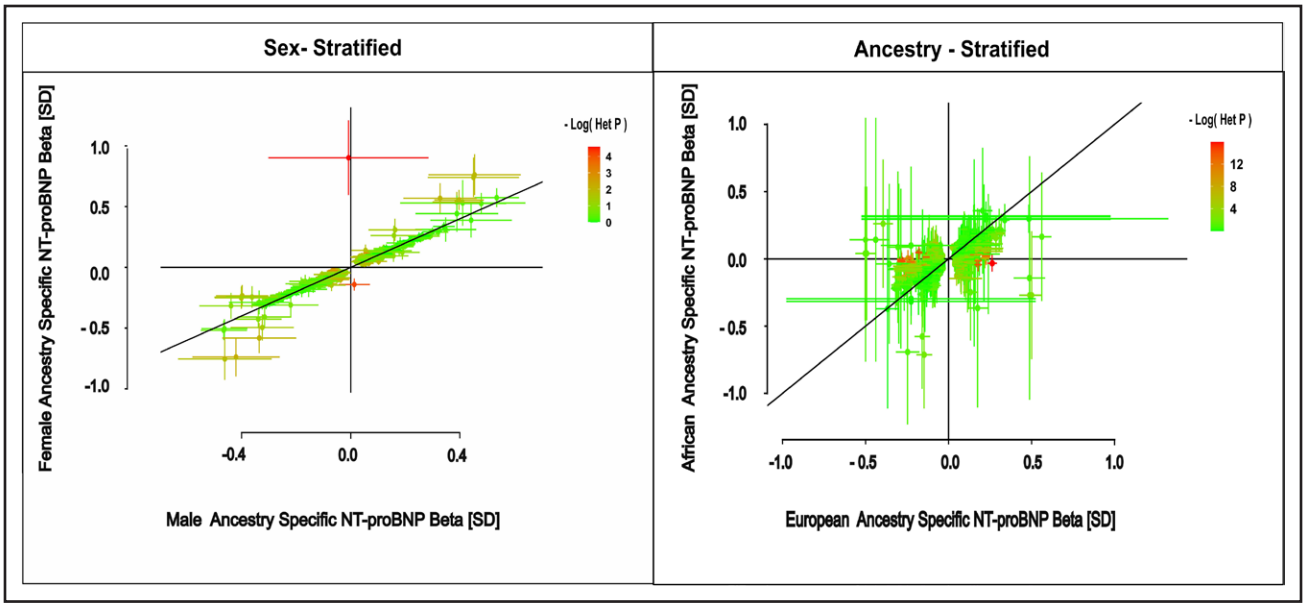

Figure 4. Heterogeneity in associations of genome-wide association study significant variants for NT-proBNP (N-terminal pro-B-type natriuretic peptide) by sex and ancestry.
Sex-based (**left**) and ancestry-based (**right**) heterogeneity in the effects of genome-wide association study significant variants for NT-proBNP has been depicted in this figure. Each variant is presented as an individual point representing the effect estimate and 95% CI in each subgroup. The dotted line represents the reference line where the effect estimates in the subgroups being compared are equal. The color of each point indicates the heterogeneity *P* value. Heterogeneity analysis was performed using meta.summaries() in the Rmeta package in R.

the association of *NPPB* with NT-proBNP levels from prior candidate gene studies and GWAS efforts for NT-proBNP.[4,11,12,64] The protein encoded by the *SLC39A8* gene belongs to the solute carrier family and is responsible for the transport of Zn, Mn, Ca, and Fe from the blood to different tissues.[68,69] The *SLC39A8* gene has been associated with the regulation of BP, which may explain its association with NT-proBNP in the current GWAS.[70]

The current study identified several novel genes associated with NT-proBNP levels, which included *SRARP* and *HSPB7* (1p36.13), *WNT8A* (5q31.2), *CDKN1A* (6p21.2), *FLNC* (7q32.1), *ZFPM2* (8q23.1), *BAG3* (10q22.2), *SYNPO2L* (10q26.11), *IGF1R* (15q26.3), and *NEDD4L* (18q21.31). Many of these proteins (*BAG3*, *FLNC*, and *SYNPO2L*) play a role in regulating muscle contraction. *BAG3*, an HSP70 chaperone, regulates a range of functions, including autophagy, late cardiac development, and maintenance of the integrity of myofibrils in cardiomyocytes.[71,72] *BAG3* mutations have been associated with the development of HF and skeletal myopathies.[73] *BAG3* knockout studies using induced pluripotent stem cell–derived cardiomyocytes demonstrate substantial myofibrillar disarray after proteasome inhibition through induction of the macroautophagy pathway.[74] Similar to *BAG3*, *NEDD4L* also regulates autophagy, additionally regulates mitochondrial function, and is involved in ubiquitination.[75] Notably, ubiquitination by *NEDD4L* is vital in regulating epithelial sodium channels.[76] Therefore, the *NEDD4L* locus may be directly associated with NT-proBNP levels through its role in sodium homeostasis or indirectly by regulation of BP.[77] The *FLNC* gene is exclusively expressed in striated muscles and encodes filamin C, which is vital for the function of actin. Filamin C crosslinks F-actin and anchors the sarcolemmal proteins to the cytoskeleton.[78] Gene-knockout murine models and human-based genetic studies have shown that *FLNC* gene mutations are associated with the development of cardiomyopathies.[78–83] *SYNPO2L* is robustly expressed in the heart and plays a role in cardiac development and the organization of the sarcomeric structure.[84] *CDKN1A* is a cell cycle inhibitor and facilitates the postnatal exit of cardiomyocytes from the cell cycle.[85] Animal studies have shown that mutations in the *CDKN1A* gene leading to increased expression of the p21 protein are associated with cardiac hypertrophy and atherosclerosis.[86,87] Although this gene was previously associated with proBNP, which is a distinct and related protein to NT-proBNP,[59] we found that it also has an association with NT-proBNP. While both analytes are biologically inactive, fragments derived from preproBNP differ in plasma stability, clearance, and clinical utility as biomarkers. Last, ZFPM2 protein is a regulator of transcription and plays an essential role in the development of the heart and coronary vasculature.[88,89]

While the associations of the abovementioned loci with NT-proBNP were novel, it is interesting to note that these loci have been associated with other cardiovascular traits in prior GWASs.[90–98] Several novel loci for NT-proBNP identified in this study, including *SYNPO2L*, *BAG3*, *NEDD4L*, and *ZFPM2*, have been associated

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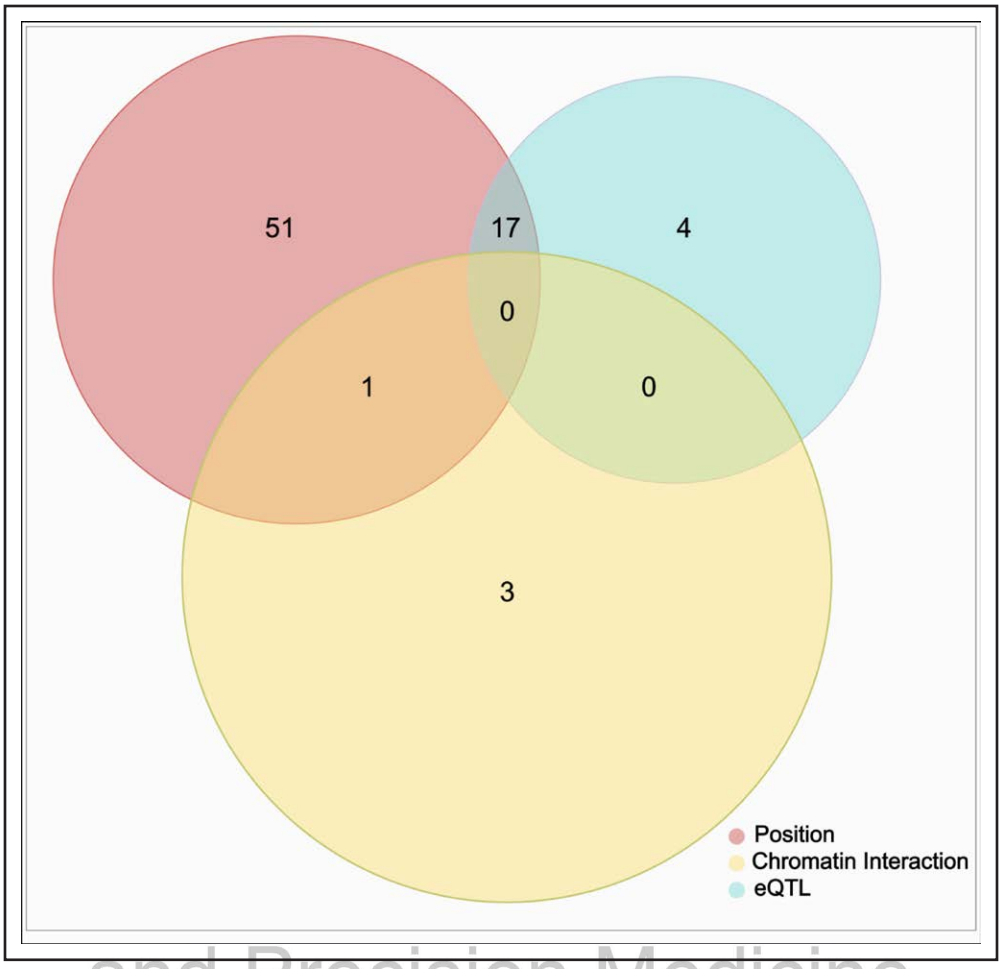

Figure 5. Venn diagram showing overlaps between genes mapped based on 3 different strategies, namely, positional mapping, expression quantitative trait loci (eQTL) mapping, and chromatin interaction mapping.
The above figure shows the overlap between genes mapped based on 3 mapping strategies, that is, positional mapping, chromatin interaction mapping, and eQTL mapping. The analysis was performed using functional mapping and annotation genome-wide association study, and genes were mapped to the lead single-nucleotide polymorphisms (SNPs) and independent SNPs that were identified as protein-coding as per the ANNOVAR annotations.

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with systolic BP.[94,96,98] Similarly, the *SYNPO2L*, *BAG3*, *NEDD4L*, *CDKN1A*, and *FLNC* loci have been associated with cardiovascular diseases such as HF (HERMES Consortium), coronary artery disease, atrial fibrillation, and dilated cardiomyopathy.[91–93,95,97]

Notably, alleles that have been implicated in HF, such as *BAG3* and *SYNPO2L*, were found to be associated with lower NT-proBNP in the current study.[73,91] While *BAG3* mutations have been shown to disrupt the sarcomeric structure and enhance cardiomyocyte autophagy,[71–73] it could be hypothesized that low NT-proBNP levels due to *BAG3* mutations may also contribute to the development of HF.

Classically, high NT-proBNP levels have been used as a gold standard to diagnose HF.[99,100] In individuals with HF, the NP levels increase in response to volume overload, but the NPs secreted in such a case are biologically inactive.[101] In healthy individuals, the biologically active hormone is secreted, which regulates a range of cardiometabolic functions, including vasodilation, glucose and lipid metabolism, and energy balance.[1–6] Low levels of bioactive NPs may lead to dysregulation of these functions, leading to an increased risk of cardiometabolic diseases such as hypertension and diabetes.[4,18,102] Also, it is important to note the close coregulation of NT-proBNP with ANP, which has been suggested to have a stronger influence on BP regulation.[2,20] ANP plays a critical role in the regulation of vascular tone and BP, and its effects on BP could contribute to the overall cardiovascular and metabolic health.[1]

Beyond coding and conventional noncoding masks, we performed rare variant aggregation analyses using the combination of the Sequence Kernel Association Test-Optimal and the Aggregated Cauchy Association Test framework, incorporating cardiac-specific promoter and enhancer regions annotated from CAGE and DNase I Hypersensitivity Site data sets. This analysis revealed significant enrichment of rare variants within

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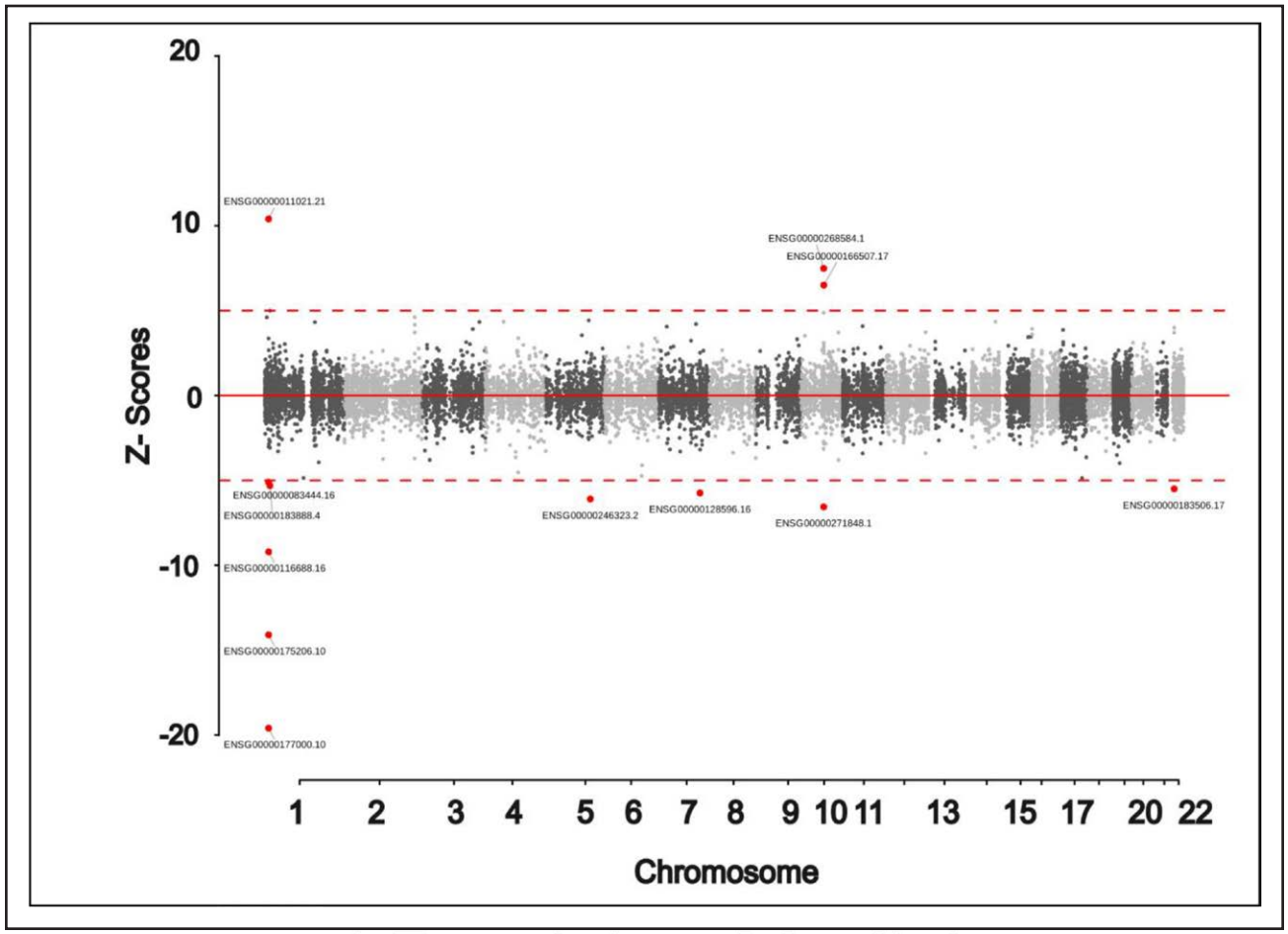

Figure 6. Transcriptome-wide association study for NT-proBNP (N-terminal pro-B-type natriuretic peptide) with left ventricle tissue–related gene expression profiles from GTEx, version 8.

The above figure shows the transcriptome-wide association study results for NT-proBNP with gene expression profiles of left ventricle tissues from GTEx, version 8. The *x* axis represents the chromosomes, and the *y* axis represents the *Z* scores. All the analyses were performed using FUSION, and a plot was generated using transcriptome-wide association study-Plotter in R. *NPPB* showed higher *Z* scores, indicating that higher expression was associated with higher NT-proBNP levels, while higher expression of *MFN2* resulted in lower NT-proBNP levels. The results are showcased in Table S13.

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regulatory elements at the *NPPB* locus, reinforcing the role of noncoding regulatory variation in modulating NT-proBNP levels. Similar enrichment was also observed at *CLCN6* and *ARHGAP42*, underscoring the broader contribution of regulatory elements. Given the observed enrichment in noncoding regulatory regions, future efforts integrating high-resolution single-cell epigenomic maps may refine regulatory variant interpretation and enhance discovery.[103–107] In addition, region-based rare variant approaches, such as sliding or adaptive window analyses, could complement gene-centric strategies. These methods have shown power to detect rare variant associations in noncoding regions, especially in large-scale whole genome sequencing studies.[108–112] Incorporating such approaches alongside tissue-specific and cell-type–specific epigenomic data may improve the discovery of regulatory variants influencing NT-proBNP levels beyond annotated gene boundaries.[113,114]

This is the first study to examine the contribution of the structural and rare variant architecture to NT-proBNP levels. SV analysis showed that NT-proBNP levels were influenced by duplications, deletions, breakends, and insertions. Rare variant analysis identified 4 masks having significant associations crossing Bonferroni correction, including 3 masks related to noncoding and regulatory elements and 1 related to coding combined mask. Combined mask of putative loss-of-function and missense, and CAGE promoter regions of *NPPB*, DNase I Hypersensitivity Site promoter regions of *CLCN6*, and CAGE Enhancer region of *ARHGAP42* were the regions identified. The SVs are not present in the *NPPB* gene, which could indicate their indirect role in regulating NT-proBNP levels either by influencing NT-proBNP synthesis by the *NPPB* gene or postsynthesis protein processing steps for NT-proBNP.

Despite leveraging fine-mapping, transcriptome-wide association study, eQTL colocalization, chromatin

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Genes	Genes Nearby 248 Independent Snps	POPs Score with Magma Zscore >= 1	Left Ventricle eQTL	Heart Atrial Appendage eQTL	Left Ventricle Chromatin Interaction	Exonic SNPs based on lead and indendent snps	RDB score <= 2 based on lead and independent snps	Gene pLI > 0.9	Genes linked to Fine-mapped Credible Snps	Gene-based tests using magma	Heart Atrial Appendage TWAS	Left Ventricle TWAS	Colocalization with PP.H4 >= 75 %	Human Ortologues with Mouse Models Evidence	Total weighted score
MFN2	1	1	0	0	0	1	1	1	0	1	1	1	0	1	9
PLOD1	1	1	0	0	0	1	1	0	0	1	1	1	0	1	8
BAG3	1	1	0	0	0	1	1	0	1	1	0	0	0	1	8
NPPA	1	1	0	0	0	1	1	0	0	1	1	0	0	0	7
NPPB	1	1	0	0	0	1	1	0	1	1	0	1	0	0	7
SLC39A8	1	1	0	0	0	0	1	0	1	0	0	1	0	1	7
HSPB7	1	1	0	0	0	1	1	0	1	1	0	1	0	0	7
C1orf167	1	1	0	0	0	1	1	0	0	1	0	0	0	0	6
KIAA2013	1	1	0	0	0	1	1	0	0	1	0	0	0	0	6
SYNPO2L	1	1	0	0	0	1	1	0	1	1	0	0	0	0	6
CLCNKA	1	1	0	0	0	0	1	0	0	1	1	0	0	0	5
P11-56N19	1	0	0	0	0	1	1	0	0	0	0	0	0	0	4
NPPA-AS1	1	0	0	0	0	1	1	0	0	0	0	0	0	0	4
MCMBP	1	1	0	0	0	1	0	1	0	0	0	0	0	0	4
ZFPM2	0	1	0	0	0	0	1	1	1	0	0	0	0	0	4
ZBTB17	1	1	0	0	0	0	0	1	0	1	0	0	0	0	4
POC1B	1	1	0	0	0	1	0	0	0	1	0	0	0	0	4
NEDD4L	1	1	0	0	0	0	0	0	1	0	0	0	0	1	4
INPP5F	1	1	0	0	0	1	0	0	0	0	0	0	0	0	3
P11-464F9	1	0	0	0	0	1	1	0	0	0	0	0	0	0	3
AGAP5	0	1	0	0	0	0	0	0	0	0	1	1	0	0	3
GALNT4	0	1	0	0	0	1	0	0	0	1	0	0	0	0	3
AGTRAP	1	1	0	0	0	0	0	0	0	0	0	0	0	0	2
VPS13D	0	1	0	0	0	0	0	1	0	0	0	0	0	0	2
MIIP	0	1	0	0	0	0	0	0	0	1	0	0	0	0	2
TXNP1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	2
P11-127H5	1	0	0	0	0	0	1	0	0	0	0	0	0	0	2
MYOZ1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	2
SPEN	0	1	0	0	0	0	0	1	0	0	0	0	0	0	2
C1B-GALN	0	0	0	0	0	1	0	0	0	1	0	0	0	0	2
P11-734K2	1	0	0	0	0	1	0	0	0	0	0	0	0	0	2
P11-981P6	1	0	0	0	0	0	1	0	0	0	0	0	0	0	2
FAM149B1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
DNAJC9	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
SWIM8-AS	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
P11-574K1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
FBXO2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
DRAXIN	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
METTL2B	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
TSPAN33	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
UBE2D3	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
P11-169K1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1
C1orf64	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
CLCNKB	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
FAM131C	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
OPN1SW	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
EPYC	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1
LUM	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1
DCN	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1

Figure 7. Prioritization of genes associated with NT-proBNP (N-terminal pro-B-type natriuretic peptide) within each locus using in silico approaches.

This figure showcases the gene prioritization of protein-coding genes identified and linked to lead single-nucleotide polymorphisms (SNPs) and independent SNPs. Genes with a prioritization score were reported in the plot above. All the lead genes are prioritized in their respective loci. *MFN2* (1p36.22) was shown to have higher evidence scores (score of 9).

interaction mapping, and rare variant analyses, the causal variants in the *NPPB* locus remain incompletely resolved. However, the rare variant analysis of regulatory regions further supports the role of *NPPB* in NT-proBNP biology. Future studies incorporating high-resolution cardiac epigenomics, allele-specific expression mapping, and functional assays will be essential to fully elucidate the regulatory mechanisms at this locus. While this study offers valuable genetic insights, further research is needed to explore how these genetic variations impact

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Table 3. Structural Variants Associated With NT-proBNP

SV ID	CHR	Start	End	Type	Length	Cytoband	Genes*	AF	β	SE	*P* value
TOPMed (N=10 416)											
DEL_1: 10483515–10485957	1	10483515	10485957	Deletion	2442	1p36.22	*PEX14*	0.003	−0.36	0.11	8.12×10^{-04}
DUP_1: 20084194–20088225	1	20084194	20088225	Duplication	4031	1p36.13	*PLA2G5*	0.003	−0.31	0.11	3.36×10^{-03}
DEL_12: 90091901–90098000	12	90091901	90098000	Deletion	6099	12q21.33	*ATP2B1* and *POC1B*†	0.328	−0.06	0.02	9.50×10^{-05}
DEL_12: 90093251–90097920	12	90093251	90097920	Deletion	4669	12q21.33		0.388	−0.04	0.01	8.68×10^{-04}
AoURP (N=3477)											
DEL 1: 10308984–10318608	1	10308984	10318608	Deletion	9624	1p36.22	*KIF1B*, *UBE4B*, and *PGD*	0.02	0.29	0.08	4.74×10^{-4}
BND 1: 10817986–10817985	1	10817986	10817985	Breakend	1		*LOC105376733* and *LOC105376735*	0.02	−0.26	0.08	2.89×10^{-3}
BND 1: 12174137–12174136	1	12174137	12174136	Breakend	1		*TNFRSF1B*, *MIR7846*, and *LOC124903844*	0.23	0.09	0.03	3.01×10^{-3}
DEL 1: 16483371–16485099	1	16483371	16485099	Deletion	1728	1p36.13	*CROCCP3*, *LINC01772*, and *RNU1-1*	0.03	−0.17	0.06	4.00×10^{-3}
BND 1: 18539949–18539948	1	18539949	18539948	Breakend	1		*KLHDC7* and *PAX7*	0.01	0.40	0.12	8.54×10^{-3}
DEL 10: 119427669–119428074	10	119427669	119428074	Deletion	405	10q26.11	*GRK5*, *MIR4681*, and *RGS10*	0.03	−0.22	0.08	4.10×10^{-3}
DUP 12: 88715257–88715394	12	88715257	88715394	Duplication	137	12q21.33	*LOC105369885* and *LOC105369886*	0.03	−0.18	0.05	2.00×10^{-3}
BND 15: 98354730–98354729	15	98354730	98354729	Breakend	1	15q26.3	*LINC02351*, *LOC105371010*, and *FAM169BP*	0.02	0.22	0.07	2.77×10^{-3}
BND 15: 100962825–100962824	15	100962825	100962824	Breakend	1		*LRRK1*, *ALDH1A3-AS1*, and *LOC105371026*	0.02	0.24	0.08	3.34×10^{-3}
INS 15: 101144527–101144526	15	101144527	101144526	Insertion	1		*LOC105371026*, *LRRK1*, and *LOC124903565*	0.03	−0.18	0.06	2.00×10^{-3}
BND 18: 57553233–57553232	18	57553233	57553232	Breakend	1	18q21.31	*FECH*, *ONECUT2*, and *NARS1*	0.43	0.09	0.03	3.13×10^{-3}
BND 4: 102350198–102350198	4	102350198	102350198	Breakend	1	4q24	*SLC39A8* and *LOC105377621*	0.003	0.59	0.18	1.00×10^{-3}
BND 6: 38374298–38374297	6	38374298	38374297	Breakend	1	6p21.2	*BTBD9*, *LOC105375045*, and *BTBD9-AS1*	0.01	−0.39	0.13	2.00×10^{-3}

SVs present in the cytoband regions of lead loci associated with inverse rank normal transformed NT-proBNP at a Bonferroni-corrected significant level ($P<0.004$) in 10 416 individuals from the TOPMed cohort and 3477 individuals from the AoURP. GWAS was conducted using the SAIGE mixed model in the Encore platform for the TOPMed cohort, and linear regression using PLINK was applied to unrelated individuals in the AoURP cohort. SVs within the regions of lead SNPs were extracted, and those meeting the Bonferroni cutoff ($P<0.004$) were reported. Eleven SVs across 3 genomic loci were identified in the TOPMed cohort, and 22 variants across 11 genomic regions were identified in the AoURP cohort. Due to the use of different SV callers for identification, we reported SVs identified from each cohort separately. AF indicates allele frequency; AoURP, All of Us Research Program; CHR, chromosome; End, structural variant end position; GWAS, genome-wide association study; NT-proBNP, N-terminal pro-B-type natriuretic peptide; SE, SE of the effect; Start, structural variant start position; SNP, single-nucleotide polymorphism; SV ID, structural variant ID; TOPMed, Trans-Omics for Precision Medicine; and β, effect for the structural variant.

*Genes that are present in SV's region. If the overlapping genes are not present, upstream/downstream genes are reported.

†Genes with overlapping regulatory elements present within each SV region.

the production, secretion, and biological activity of NT-proBNP at a functional level. The potential for gene-editing techniques, such as clustered regularly interspaced short palindromic repeats, to target specific genetic variants influencing NT-proBNP levels offers a promising avenue for future therapeutic developments for HTN and HF. In addition, clinical trials assessing the impact of modulating NT-proBNP through pharmacological or gene therapy approaches will be crucial in translating these findings into clinical practice.

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Table 4. Gene-Level Aggregates Associated With NT-proBNP

Gene	Mask	Variants included	Chr	Gene start	Gene end	β	SE	*P* value	Direction	Cytoband
NPPB	pLoF+missense	All	Chr1	11857464	11858931	0.33	0.02	4.24×10^{-65}	+−++	1p36.22
	Missense	All	Chr1	11857464	11858931	0.33	0.02	3.52×10^{-65}	+−++	1p36.22
	Promoter (CAGE)	All	Chr1	11857464	11858931	0.20	0.01	2.33×10^{-44}	+−++	1p36.22
CLCN6	Promoter (DHS)	All	Chr1	11806096	11848079	0.08	0.02	1.34×10^{-07}	++++	1p36.22
KIAA1257	Missense	All	Chr3	128909866	129002690	−0.06	0.01	3.34×10^{-07}	——	3q21.3
ARHGAP42	Enhancer (CAGE)	All	Chr11	100687653	100993941	−0.13	0.02	9.2×10^{-08}	?——	11q22.1
RNF151	Intronic	All	Chr16	1966823	1968975	−0.09	0.02	8.83×10^{-07}	——	16p13.3

Rare variant associations across coding, noncoding, and regulatory element masks with inverse rank normal transformed NT-proBNP levels were evaluated in 81 213 individuals from 4 cohorts: TOPMed, AoURP, UK Biobank, and the REGARDS study. Rare variant aggregation testing was performed using the SKAT-O-ACAT model implemented in Regenie v3.4.1. Coding-based masks included pLoF, missense, and splice site variants; noncoding masks included intronic, intergenic, downstream, upstream, and UTR variants; and regulatory element masks were defined using cardiac-specific promoter and enhancer regions annotated from CAGE and DHS data sets. Meta-analyses of gene-level association results were conducted using METAL. Six masks across 5 genes surpassed the Bonferroni-corrected significance threshold ($P\leq2.71\times10^{-6}$), including signals from regulatory regions. Cytogenetic band annotations for the reported genes were derived using in-house tools to support genomic localization. ACAT indicates Aggregated Cauchy Association Test; CAGE, Cap Analysis of Gene Expression; Chr, chromosome where the gene is located; Cytoband, cytogenetic band representing the chromosomal region where the gene is located; DHS, DNase I Hypersensitivity Site; Direction, direction of effect in each individual cohort included in the meta-analysis; Gene end, genomic coordinate for the gene end (GRCh38); Gene start, genomic coordinate for the gene start (GRCh38); Gene, official gene symbol; Mask, rare variant aggregation mask used for gene-based testing (eg, pLoF, missense, promoter, and enhancer); NT-proBNP, N-terminal pro-B-type natriuretic peptide; *P*, *P* value from the meta-analysis; pLoF, putative loss-of-function; SE, SE of the effect size; SKAT-O, Sequence Kernel Association Test-Optimal; Variants included, total number of rare variants aggregated in the mask; and β, effect size estimate for the association between the aggregated rare variants and log-transformed standardized NT-proBNP levels.

The major strength of the current study is the large sample size of this study. The study meta-analyzed data from the TOPMed cohorts, UK Biobank, REGARDS study, and the AoURP, which provided a cohort of individuals from diverse ancestral backgrounds. The large sample size, coupled with the diverse ancestral background, boosted genetic discovery. However, this study has several limitations. First, the method of measuring NT-proBNP varied between the UK Biobank and other cohorts included in this study. NT-proBNP was measured using the Olink PEA in the UK Biobank, while the conventional ELISA assay was utilized to measure NT-proBNP in the other cohorts. The current study used inverse rank-normalized NT-proBNP values across the cohorts, which minimizes the differences in the NT-proBNP levels due to measurement on different scales. Furthermore, NT-proBNP measured on ELISA and PEA has been shown to have a strong correlation across the spectrum of health.[115] Second, genotyping techniques varied across the cohorts. The AoURP and TOPMed cohorts utilized WGS, while the UK Biobank and REGARDS study used genotyping arrays. This limitation was mitigated by imputing the genotyping arrays with multiancestry panels such as the TOPMed imputation panel, ensuring consistent genetic data sets. Third, because the majority of the data set consists of individuals of European ancestry, our analysis may be underpowered to detect signals across multiple ancestry groups. Although the direction of effects appears consistent, genes that were significant in one ancestry group did not reach statistical significance in other groups. A similar pattern was observed in the sex-stratified analysis. Fourth, the exclusion of prevalent HF events based on *International Classification of Diseases* codes and NT-proBNP guidelines was implemented to focus on a healthier population and to avoid confounding by existing heart conditions. However, this exclusion limits our ability to draw causal inferences about the relationship between NT-proBNP variants and incident HF.

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CONCLUSIONS

This study focused on elucidating the genetic determinants of NP levels, thereby providing valuable insights into the underlying genetic architecture of these cardiometabolic hormones. By identifying variants associated with NP levels, such as those observed at the *SLC39A8* locus linked to high BP, the *FLNC* locus implicated in cardiomyopathy, and the *BAG3* locus associated with HF, this study not only contributes to our understanding of the genetic basis of NP system regulation but also unveils novel biological pathways relevant to cardiovascular disease pathogenesis. These findings offer promising avenues for further investigation into the molecular mechanisms underlying cardiovascular diseases, which may pave the way for the development of targeted interventions and precision medicine approaches in cardiovascular care.

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Supplemental Material

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